

IPL CRICKET ANALYSIS

This report documents an Exploratory Data Analysis (EDA) project performed on the Indian Premier League (IPL) dataset. It is intentionally written as an analytics report because the notebook does not implement predictive machine learning models. The report structure follows the style of the provided Uber Ride Analysis sample while remaining technically accurate.

Project Overview

The objective is to analyze IPL match- and ball-level datasets to understand batting performance, bowling performance, player achievements, team performance, season-wise trends, and match outcomes using Python, Pandas, Matplotlib, Seaborn and Plotly.

1.Dataset Description

Match Dataset (matches.csv)

The matches.csv dataset contains match-level information for every IPL game. Each record represents a single match and includes information about the competing teams, venue, toss details, match winner, player of the match, and season.

Some of the important attributes present in this dataset are:

Column Name	Description
Id	- Unique Match ID
Season	- IPL Season
City	- Match Venue City
Date	- Match Date
team1	- First Team
team2	- Second Team
toss_winner	- Team Winning Toss
toss_decision	- Bat or Field
Winner	- Winning Team
player_of_match	- Best Player Award

Delivery Dataset (deliveries.csv)

The **deliveries.csv** dataset contains ball-by-ball information for every IPL match. Every row represents a single delivery bowled during the tournament. This dataset enables detailed batting and bowling analysis by recording runs scored, wickets taken, extras, and dismissal details.

Important columns used in this project include:

Column Name	Description
match_id	Match Identifier
inning	Innings Number
over	Over Number
ball	Ball Number
batting_team	Batting Team
bowling_team	Bowling Team
batsman	Batsman Name
bowler	Bowler Name
batsman_runs	Runs Scored by Batsman
total_runs	Total Runs on Ball
dismissal_kind	Type of Wicket
player_dismissed	Dismissed Player

2.Data Preprocessing

Data preprocessing is an important stage of the analysis process because raw datasets often contain missing values, inconsistent information, and redundant records. Proper preprocessing improves data quality and ensures accurate analytical results.

The following preprocessing steps were carried out in this project.

Importing the Dataset

Initially, the IPL datasets were imported into Jupyter Notebook using the Pandas library. The datasets were stored as DataFrames for efficient processing and analysis.

Data Inspection

The structure of the datasets was examined using functions such as `head()`, `info()`, and `describe()` to understand the available attributes, data types, and statistical properties.

Handling Missing Values

The datasets were inspected for missing values. Missing or inconsistent values were identified and handled appropriately to avoid incorrect statistical analysis. Columns containing dismissal information naturally contained missing values for deliveries where no wicket occurred, and these values were treated accordingly during wicket analysis.

Data Cleaning

The datasets were cleaned by correcting inconsistent column names, removing unnecessary records where required, and ensuring that categorical values were represented consistently. This step helped maintain data integrity throughout the analysis.

Merging Datasets

The match dataset and delivery dataset were merged using the common match identifier. This integration enabled season-wise player analysis and allowed ball-level statistics to be associated with corresponding match information.

Feature Engineering

Several additional analytical metrics were generated from the raw data, including:

- Total Runs scored by batsmen
- Number of Fours

- Number of Sixes
- Strike Rate
- Total Wickets
- Bowling Economy
- Orange Cap Statistics
- Purple Cap Statistics
- Season-wise Runs
- Team Performance Metrics

Data Aggregation

Grouping and aggregation techniques were extensively used to summarize player and team statistics. Functions such as `groupby()`, `sum()`, `count()`, `mean()`, and sorting operations helped identify top-performing players and teams across different seasons.

Methodology

This project follows an Exploratory Data Analysis (EDA) methodology rather than predictive modeling. The overall workflow is illustrated below:

1. Dataset Collection
2. Data Loading
3. Data Inspection
4. Data Cleaning
5. Data Integration
6. Feature Engineering
7. Exploratory Data Analysis
8. Visualization
9. Insight Generation
10. Conclusion

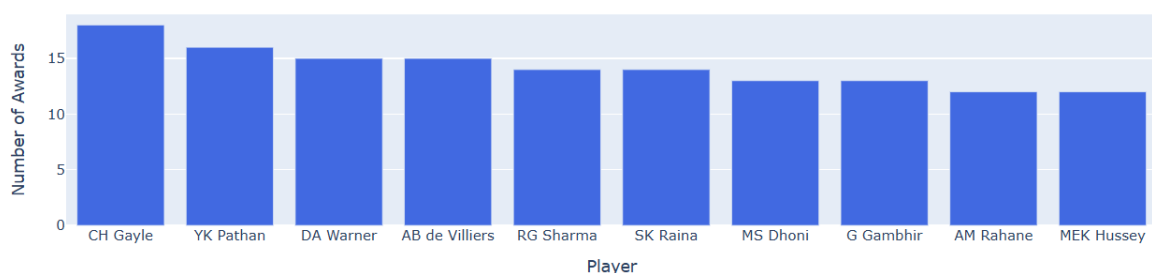
3.EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is the process of examining datasets to understand their characteristics through statistical summaries and visualizations. In this project, EDA was performed on IPL match-level and ball-by-ball datasets to identify trends related to team performance, batting achievements, bowling performance, and season-wise statistics. The analysis was carried out using Python libraries such as Pandas, Matplotlib, Seaborn, and Plotly. Each visualization presented in this chapter highlights a specific aspect of IPL performance. The observations and insights obtained from these analyses help in understanding player consistency, team strategies, and tournament trends.

3.1 Player of the Match Analysis

The objective of this analysis is to identify the players who have received the highest number of **Player of the Match** awards across IPL seasons. These awards recognize exceptional individual performances that significantly contribute to a team's victory.

Top 10 Player of the Match Winners



Observation:

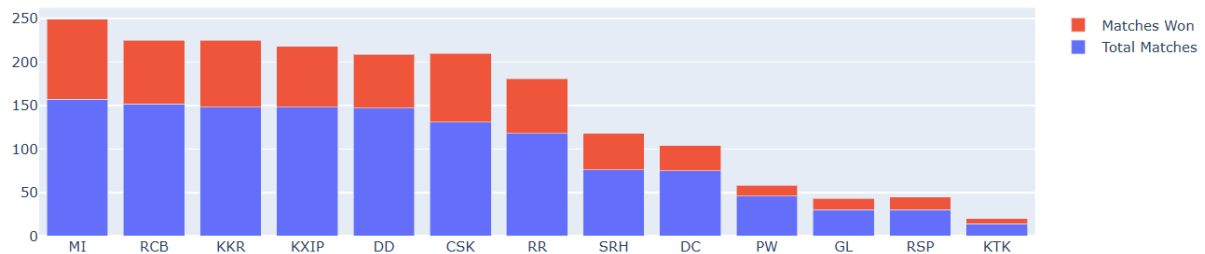
The visualization displays the number of Player of the Match awards won by different players. A small group of players has received the award multiple times, indicating consistent match-winning performances over several IPL seasons. Most players have received the award only once or twice, while a few exceptional performers dominate the chart.

Key Insight:

The Player of the Match analysis highlights the importance of consistent individual performances in determining match results and identifies players who have had a lasting impact on the IPL.

3.2 Teams Played Maximum Matches

This analysis aims to compare the number of matches played by each IPL team throughout the tournament.



Observation:

The graph shows that franchises participating in the IPL for a longer duration have naturally played a greater number of matches. Teams introduced in later seasons have comparatively lower match counts.

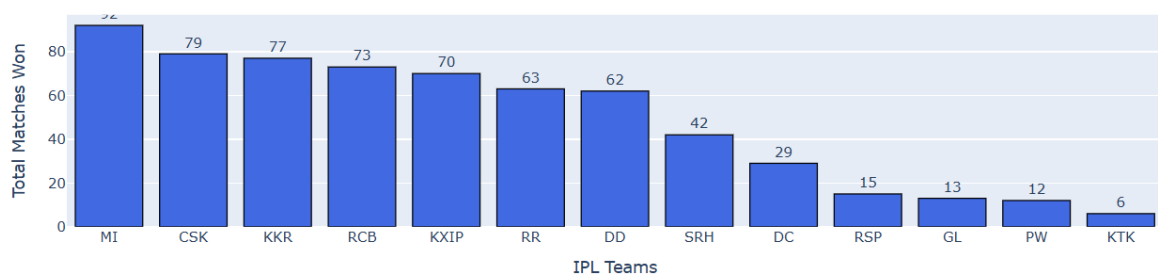
Key Insight:

This analysis provides an overview of team participation and forms the basis for comparing team performances across IPL history.

3.3 Team Winning Analysis

The objective is to compare the total number of matches won by different IPL teams.

Team Winning Analysis in IPL



Observation:

The graph illustrates the total victories achieved by each IPL franchise. Some teams consistently recorded higher numbers of wins, indicating sustained performance across seasons.

Key Insight:

Teams with consistently higher win counts generally maintained balanced batting and bowling units, contributing to their long-term success.

3.4 Season-wise Total Runs

This analysis compares the total runs scored in each IPL season.



Observation:

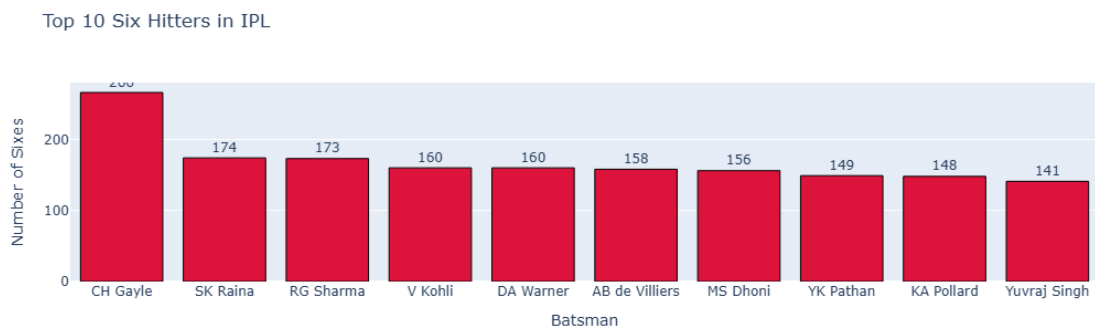
The total runs scored vary across different IPL seasons. Some seasons recorded higher overall scores due to aggressive batting performances and improved strike rates.

Key Insight:

Season-wise run analysis helps identify the evolution of batting approaches and demonstrates how IPL has become increasingly high-scoring over time.

3.5 Top Six Hitters

To identify aggressive batsmen based on the number of sixes scored.



Observation:

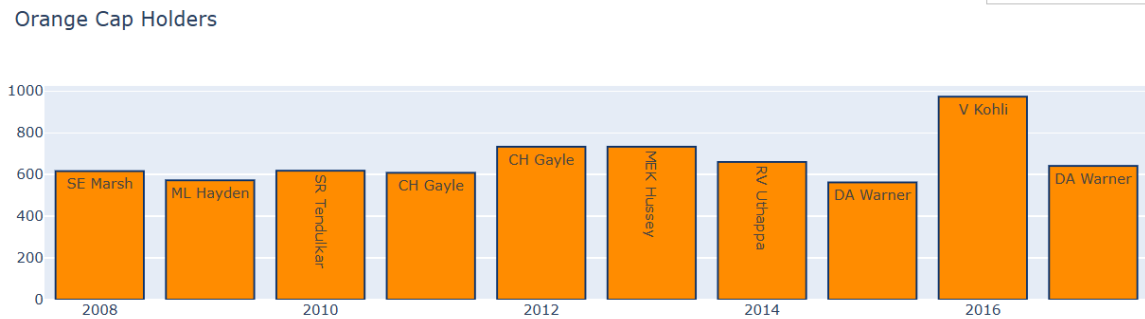
The graph highlights power hitters capable of clearing the boundary regularly. These players often contribute significantly during the middle and death overs.

Key Insight:

Aggressive batsmen with high six counts play an important role in setting challenging targets and successfully chasing totals.

3.7 Orange Cap Analysis

The objective of this analysis is to identify the highest run scorer (Orange Cap winner) in each IPL season.



Observation:

The visualization highlights the leading run scorer for every IPL season. Orange Cap winners consistently accumulated a large number of runs through excellent batting performances across multiple matches.

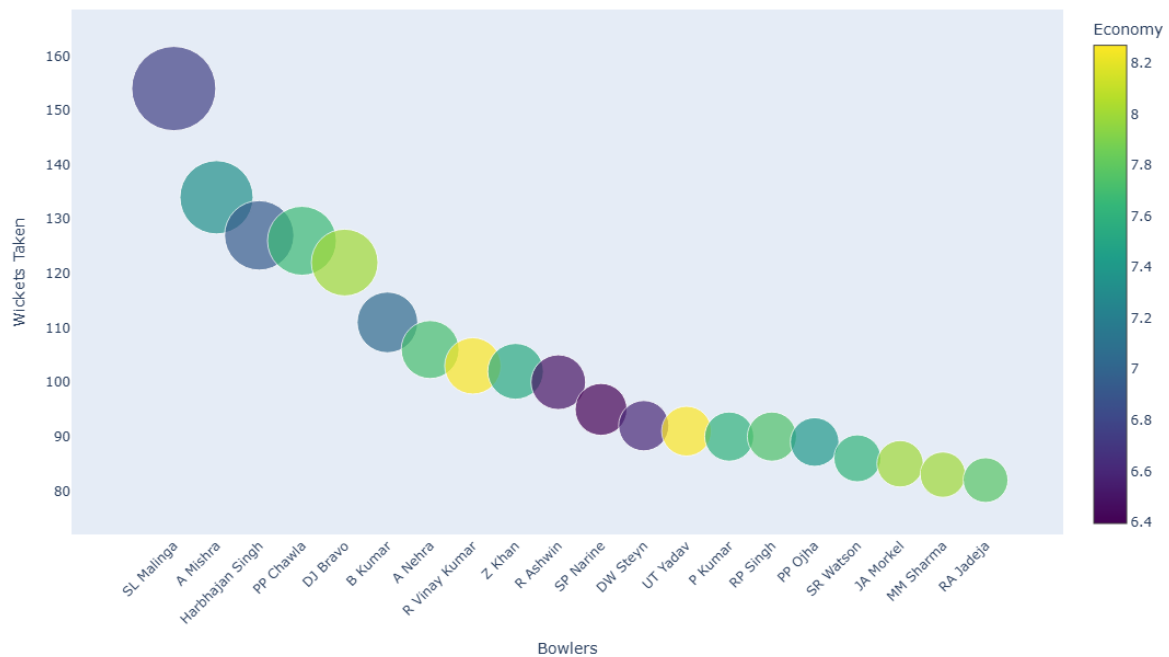
Key Insight:

Orange Cap analysis demonstrates that consistent batting performances across an entire season contribute more to success than individual high-scoring innings.

3.8 Top Bowlers Analysis

The objective is to identify bowlers who have taken the highest number of wickets in the IPL.

Top 20 Wicket Taking Bowlers



Observation:

The visualization displays bowlers with the highest wicket counts. Successful bowlers consistently dismissed batsmen across different IPL seasons and contributed significantly to their teams.

Key Insight:

Consistent wicket-taking ability is a key characteristic of successful T20 bowlers and often determines a team's defensive strength.

3.9 Purple Cap Analysis

The objective is to identify the highest wicket-taker in each IPL season.

Purple Cap Holders



Observation:

The graph displays the bowlers who secured the highest number of wickets in each IPL season. These bowlers consistently delivered match-winning performances throughout the tournament.

Key Insight:

The analysis emphasizes the importance of wicket-taking bowlers in restricting opposition teams and influencing match results.

4.KEY FINDINGS AND RESULTS

The Exploratory Data Analysis performed on the IPL datasets revealed several valuable insights regarding player performances, team achievements, batting efficiency, bowling effectiveness, and season-wise variations. The major findings of the project are summarized below.

4.1 Batting Performance

The batting analysis revealed that a small group of batsmen consistently dominated IPL scoring charts. These players accumulated a large number of runs over multiple seasons while maintaining impressive strike rates. The boundary analysis further showed that successful batsmen balanced aggressive stroke play with consistent run accumulation. Players who scored a high number of fours demonstrated excellent timing and placement, whereas players with more sixes displayed aggressive batting suitable for T20 cricket. Overall, batting statistics indicated that consistency across an entire season plays a more significant role than individual high-scoring innings.

4.2 Bowling Performance

The bowling analysis identified several bowlers who consistently contributed by taking wickets while maintaining economical bowling figures. The Purple Cap analysis highlighted bowlers who dominated throughout individual IPL seasons. The economy rate analysis demonstrated that some bowlers were successful in restricting opposition scoring even under pressure. Such bowlers proved valuable during powerplay and death overs where controlling runs becomes highly challenging. The dismissal analysis further revealed that catches and bowled dismissals accounted for the majority of wickets, indicating the effectiveness of pace bowling and field placements.

4.3 Team Performance

The team analysis showed considerable variation in performances among IPL franchises. Certain teams consistently achieved higher win counts due to balanced batting and bowling units. Long-established franchises naturally recorded more matches because of their continued participation throughout the tournament.

The head-to-head comparison illustrated historical rivalries between teams and highlighted performance differences across seasons.

4.4 Season-wise Trends

Season-wise analysis demonstrated noticeable changes in scoring patterns across IPL history. Modern IPL seasons generally witnessed higher run totals due to aggressive batting approaches, improved strike rates, and evolving match strategies. The increasing emphasis on attacking cricket has significantly influenced scoring trends throughout the tournament.

4.5 Player Achievements

Individual player analysis highlighted several consistent performers through:

- Player of the Match awards
- Orange Cap performances
- Purple Cap performances

These awards reflect exceptional consistency and match-winning contributions over multiple seasons.

5.CHALLENGES FACED

During the execution of this project, several practical challenges were encountered while processing and analyzing the IPL datasets. Addressing these challenges was essential for obtaining reliable and meaningful analytical results.

5.1 Large Dataset Size

The IPL ball-by-ball dataset contains thousands of records covering multiple seasons. Processing such a large dataset required efficient data manipulation techniques using Pandas.

5.2 Missing Values

Certain columns, particularly those related to player dismissals, naturally contained missing values because wickets do not occur on every delivery. These missing values were handled appropriately during analysis.

5.3 Data Integration

The project required combining match-level information with ball-by-ball data using common match identifiers. Ensuring correct dataset integration was necessary for performing season-wise player analysis.

5.4 Statistical Aggregation

Calculating strike rates, economy rates, wickets, season-wise runs, and player rankings involved multiple aggregation operations. Proper grouping and filtering techniques were required to generate accurate statistics.

5.5 Visualization

Presenting complex cricket statistics in an understandable manner required careful selection of appropriate charts and visualization techniques.

6.FUTURE SCOPE

Although this project focuses entirely on Exploratory Data Analysis, there are several opportunities for extending the work in future.

Some possible future enhancements include:

- Match Winner Prediction using Machine Learning
- Player Performance Prediction
- Fantasy Cricket Recommendation System
- Venue-wise Performance Analysis
- Team Selection Optimization
- Toss Impact Prediction
- Live Match Analytics Dashboard
- Deep Learning-based Cricket Performance Analysis

These enhancements can further improve decision-making by providing predictive and prescriptive insights in addition to descriptive analytics.

CONCLUSION

The IPL Data Analysis project successfully applied Exploratory Data Analysis (EDA) techniques to analyze historical IPL datasets. Using Python and libraries such as Pandas, Matplotlib, Seaborn, and Plotly, the project examined batting performance, bowling statistics, team performance, and season-wise trends through meaningful visualizations.

The analysis provided valuable insights into player consistency, team achievements, and overall tournament performance. Since the project focuses on EDA, no predictive machine learning models were implemented. Overall, the project achieved its objectives by transforming raw cricket data into useful insights, demonstrating the importance of data analytics in sports.